

ASAF BELILUS

Full-Stack Software Engineer · Spring Boot, React & PostgreSQL

CONTACT

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[LinkedIn](#)
[GitHub](#)
[Portfolio](#)

SKILLS & TECHNOLOGIES

Languages

Java, Python, TypeScript, C/C++

Frontend

React, Vite, i18n/RTL (multi-language)

Backend

Spring Boot, REST APIs, JWT Auth

Data

PostgreSQL, versioned schema migrations

Tools & Testing

Docker, Git, automated testing

EDUCATION

M.Sc. Computer Science

Ben-Gurion University of the Negev
Researcher, Prof. Gera Weiss's Lab · 2026 – Present (Expected Oct 2027)
Current GPA 95.

B.Sc. Computer Science

Ben-Gurion University of the Negev · 2022 – 2025
Final GPA 84.2 · Active swimmer, BGU ASA competitive swim team.
Coursework: Operating Systems, Computer Architecture, Systems Programming, Databases

ACTIVITIES

Teaching Assistant — BGU · 2026 – Present

Supervises practicum coursework and grades/examines student papers.

Individual/Group Swimming Instructor — Wingate Institute · 2014 – Present

Teaches fundamentals and advanced technique to swimmers of all ages.

Volunteer Math & CS Tutor — BGU · 2026 – Present

Tutors first-year students in math and CS, focused on reservist students returning to their studies.

LANGUAGES

Hebrew — Native
English — Fluent

PROFILE

Computer Science M.Sc. student seeking a full-stack software engineering student position. Designed and built SwimEdge end to end with Spring Boot, React, and PostgreSQL: six server-enforced roles, 23 Flyway migrations, and 811 backend and frontend tests. Works from domain requirements and data modeling through APIs, multilingual UI, and verification.

RESEARCH

Swimmer Biomechanics Research — Prof. Gera Weiss's Lab

Ben-Gurion University of the Negev · in collaboration with Prof. Raziell Reimer, Faculty of Engineering · 2026 – Present

- Conducting cross-disciplinary research applying computer vision and control theory to competitive-swimming biomechanics, bridging the Computer Science and Engineering faculties.
- Built Python pipelines that reconstruct swimmer joint motion from 3-D underwater pose-estimation data, using forward/inverse kinematics and nonlinear least-squares optimization to drive a third-party hydrodynamic simulation engine.
- Diagnosed systematic model-fitting errors (degenerate optimization bounds, skeletal geometry mismatches) via quantitative probing against ground-truth motion-capture data, backed by automated regression testing.

FEATURED PROJECT

SwimEdge — Full-Stack Competitive Swimming Platform

Founder & Sole Engineer · 2025 – Present · [GitHub](#)

- Designed and built a full-stack platform for competition management and performance analytics, motivated by a gap identified firsthand as a competitive swimmer — owning architecture, backend, frontend, and infrastructure end to end.
- Engineered a modular, service-oriented Spring Boot + PostgreSQL backend and an interactive React/Vite frontend; evolved the schema through 23 versioned migrations with continuously verified automated test suites.
- Built a custom parser converting official Israeli Swimming Association PDF/Excel results into clean relational data and built a multi-language RTL-ready interface. Opened direct discussions with the Israel Swimming Association and engaged Israel's national sports administration (Ministry of Culture and Sport) on the opportunity for technological modernization across sport federations.

MILITARY SERVICE

Combat Commander, Paratrooper Brigade, IDF

2016 – 2019

- Trained and mentored newly recruited soldiers; led operations and managed company-level logistics.

Reservist, Paratrooper Brigade, IDF

2019 – Present

- Lead frontline teams under pressure, maintaining calm and accountability.